SUPPLEMENTARY MATERIAL

Supplementary material for: Evaluation design can decide the winner: spatial and acquisition leakage in few-label hyperspectral classification

ARTICLE HISTORY
Compiled November 1, 2026

Table S1. Average accuracy (AA, %) / Cohen's κ on spatially disjoint test areas (mean over 5 seeds).

Scene	Labelled	LightGBM	Spectral tr.	Grouped-band tr.	Conv-stem tr.	HybridSN
Indian Pines	1%	**36.5** / **0.28**	29.2 / 0.23	31.7 / 0.26	20.7 / 0.19	31.6 / 0.25
	5%	**42.9** / **0.39**	34.6 / 0.28	33.2 / 0.26	28.7 / 0.22	38.4 / 0.33
	10%	**44.6** / **0.39**	40.5 / 0.33	39.7 / 0.31	32.4 / 0.27	40.3 / 0.34
	20%	47.3 / 0.41	**47.7** / **0.44**	44.9 / 0.40	37.0 / 0.31	43.0 / 0.36
Pavia Univ.	0.5%	64.6 / 0.58	61.9 / 0.55	58.5 / 0.53	53.8 / 0.59	**71.3** / **0.69**
	1%	68.2 / 0.61	64.6 / 0.57	63.9 / 0.57	68.6 / 0.68	**80.0** / **0.74**
	2%	73.5 / 0.68	73.9 / 0.66	70.3 / 0.60	80.9 / **0.78**	**82.8** / **0.78**
	5%	79.7 / 0.74	82.0 / 0.75	77.5 / 0.69	86.6 / 0.81	**89.1** / **0.85**
Salinas	0.5%	**79.3** / 0.70	68.9 / 0.61	73.7 / 0.66	65.9 / 0.66	79.0 / **0.73**
	1%	82.9 / 0.75	80.3 / 0.69	82.8 / 0.72	79.4 / 0.77	**89.8** / **0.84**
	2%	84.9 / 0.78	85.2 / 0.75	85.9 / 0.75	85.3 / 0.81	**91.4** / **0.85**
	5%	87.0 / 0.79	88.0 / 0.79	88.4 / 0.79	90.2 / 0.85	**93.0** / **0.88**

The best value per row is in bold.

Table S2. Overall accuracy (%) under a random pixel split (mean ± SD over 5 seeds); compare Table 2 of the letter.

Scene	Labelled	LightGBM	Spectral tr.	Grouped-band tr.	Conv-stem tr.	HybridSN
Pavia Univ.	0.5%	76.15 ± 1.65	74.44 ± 1.65^{a}	74.33 ± 0.99^{a}	72.75 ± 4.47	$\mathbf{83.94 \pm 2.13}^{a}$
	1%	81.35 ± 0.26	78.98 ± 1.52	77.18 ± 0.73^{a}	85.62 ± 2.29^{a}	$\mathbf{89.84 \pm 1.56}^{a}$
	2%	85.94 ± 0.76	83.54 ± 2.49	79.91 ± 0.53^{a}	92.33 ± 0.75^{a}	$\mathbf{95.68 \pm 1.20}^{a}$
	5%	89.93 ± 0.27	90.64 ± 1.39	86.44 ± 1.88^{a}	97.81 ± 0.61^{a}	$\mathbf{98.97 \pm 0.39}^{a}$
Salinas	0.5%	82.45 ± 1.69	74.55 ± 2.10^{a}	77.23 ± 0.93^{a}	76.26 ± 4.93	$\mathbf{83.48 \pm 2.23}$
	1%	86.06 ± 0.96	81.02 ± 1.21^{a}	82.69 ± 1.19^{a}	88.30 ± 1.18	$\mathbf{93.42 \pm 2.15}^{a}$
	2%	88.61 ± 0.40	85.77 ± 0.72^{a}	85.14 ± 0.42^{a}	93.82 ± 0.36^{a}	$\mathbf{97.15 \pm 0.86}^{a}$
	5%	90.92 ± 0.18	89.09 ± 0.55^{a}	88.70 ± 0.72^{a}	97.20 ± 0.24^{a}	$\mathbf{99.49 \pm 0.07}^{a}$

Test sets contain 37 638–39 563 (Pavia University) and 48 423–50 859 (Salinas) pixels. [a]Paired difference from LightGBM significant ($q < 0.05$, same family as Table 1 of the letter).